READ THIS FIRST

# An automated buyer, not a price bot

What we built, what thirteen real practice negotiations revealed, and what to improve next.

## The idea in one minute

Imagine a pump manufacturer buying metal parts. A supplier asks for a higher price. Our software reads the quote, checks what the factory needs, proposes a trade, reads the reply, and either agrees a safe deal or walks away. It compares the **whole cost of supplying the factory**, not just the number printed next to one part.

| Experiment | Average /100 | What this tells us |
|---|---|---|
| All 10 hosted problems | **50.30** | Broad baseline: one attempt at every available problem, same code. |
| 3 random follow-ups | **39.02** | After reflection and fixes; compare with the same three earlier cases. |
| Same 3 before fixes | 38.44 | The fairer comparison, though still only one observation per version. |

### The system is connected. Its performance is uneven.

All 13 sessions produced valid deals. The architecture is working end to end, but that is not the same as consistently finding a strong deal. The full set averages 50.30, with especially weak index-linked cases. Some financial effects remain unobservable, and idle RFQ threads can time out.

## What changed during this task

We finished bounded clarification and offer recovery, ran all ten cases, read every conversation, then added a request to restore a better earlier quote, stricter invitation recognition, evidence answers during questions, and a fresh server-state check before closing. Final local gate: **305 passed, 1 skipped, 15 subtests passed**.

Reading guide: pages 2-7 explain the system; pages 8-10 explain the experiment and reflection; pages 11-12 support the next decision and reproduce the evidence. No merge, commit, push or default-agent change was made.

# What are we trying to win?

The supplier is simulated. The negotiation and accepted contract are real API interactions.

## Our role

The fictional buyer is Hollerau Pumpenwerke, a German pump maker. A case gives us a part, required quantity, purchasing period, hard requirements and an alternative. One to three supplier bots respond by email. They can move prices and terms, refuse requests or stop negotiating. A **must-have** is a hard requirement, such as delivery within 42 days, that a lower price cannot override.

## What counts as a deal

The structured term sheet is binding; email prose alone is not. We must accept the supplier's **current primary offer ID**. A cheaper alternative mentioned in a reply must first be confirmed as that standing offer. Missing terms may fall back to the supplier's quote PDF, not our preferred defaults.

### Score = share of the available whole-cost saving

**(opening cost - accepted cost) / (opening cost - hidden best cost)**, limited to 0-100%. A broken must-have scores zero. No deal takes the alternative's score, or zero without an alternative. A valid but unchanged opening offer can still score zero.

| Simple example from the brief | Whole cost |
|---|---|
| Opening: EUR 43.20, payment after 30 days | EUR 347,904 |
| Deal: EUR 41.60, payment after 60 days | EUR 332,800 |
| Illustrative hidden best | EUR 324,224 |
| Saving 15,104 / available saving 23,680 | 63.78 /100 |

## Constraints that shape the software

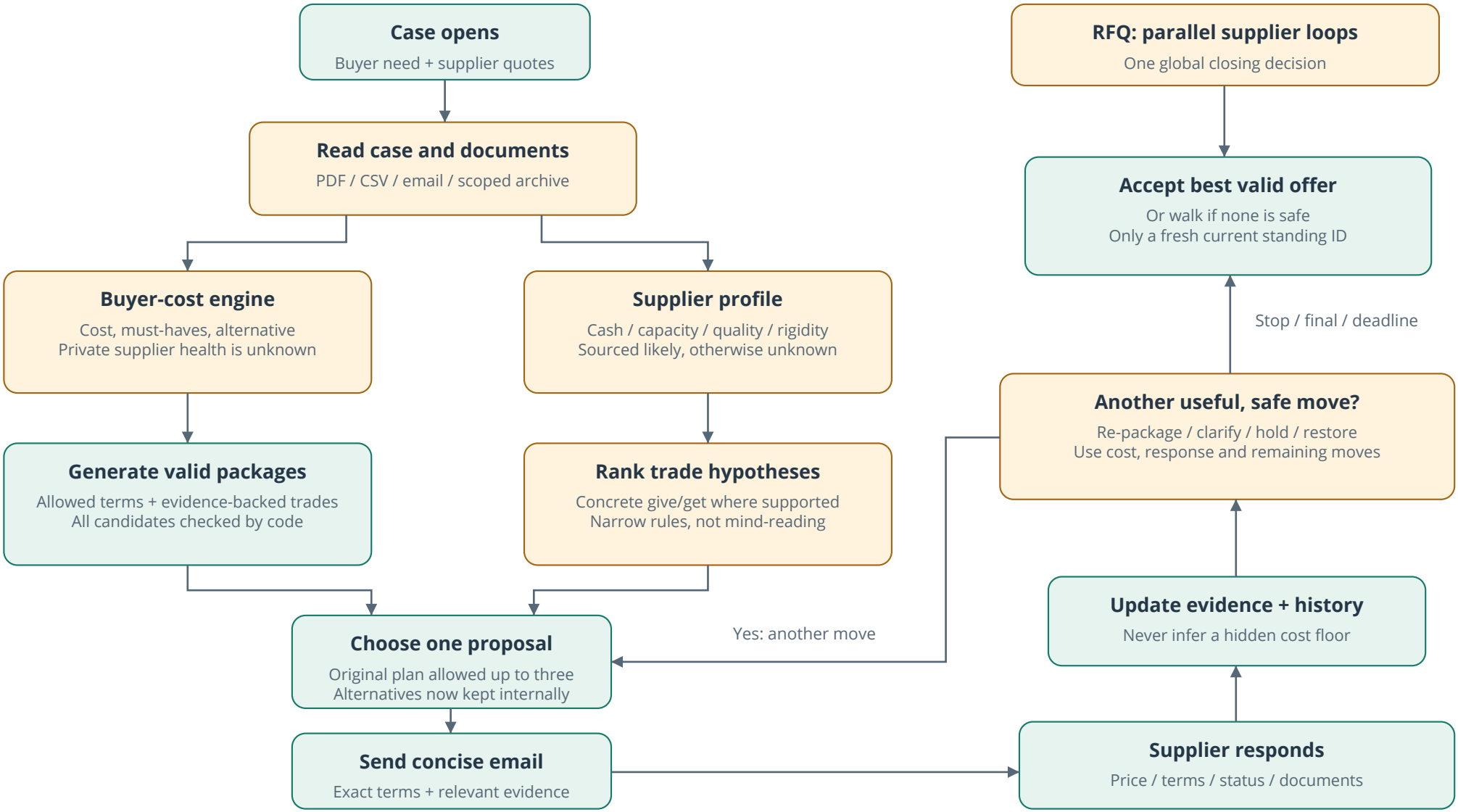
Up to three complete packages per email, ten rounds by default, and 60 seconds to respond after each supplier reply. The final has five waves with an eight-minute cap each. The runner must be unattended; it must not read final leaderboards or event feeds. The score rewards cost, not the number of citations or the length of our messages.

Rules source: bundled brief/brief.md, brief/api.md and brief/submission.md. Server-provided practice quota is authoritative; this benchmark observed a 450-session limit, while the bundled brief says 200.

ARCHITECTURE MAP

# The original loop, now connected

Green = implemented and exercised. Amber = implemented with important limits. Arrows show the running flow, not a new proposed architecture.



COMPONENTS 1-3

# Read facts without inventing motives

The profile is a small, source-linked set of hypotheses, not a psychological model.

| Input | What it is used for | What it must not become |
|---|---|---|
| Supplier emails, quote PDFs, CSVs and relevant supplier letters | Quoted terms and explicit supplier statements, with source IDs. | A guessed cost floor, financing rate or unstated guarantee. |
| Buyer case and published rates | Need, hard limits, permitted terms, total cost and reservation. | Evidence that the supplier needs cash or has spare capacity. |
| Eligible buyer purchase orders / reference quotes | An honest price-history argument when relevant and requested. | Proof of today's supplier cost or identical commercial terms. |

| Profile dimension | Supported example | Policy response |
|---|---|---|
| Cash sensitive | Supplier explicitly requests earlier payment. | Offer earlier payment only for a lower proposed price. |
| Capacity sensitive | Supplier asks for binding volume / commitment. | Test commitment within buyer need and horizon. |
| Quality sensitive | Supplier provides a numeric quality guarantee. PPM means defects per million units. | Use permitted guaranteed terms; never invent a stricter promise. |
| Price rigid | Sourced fixed-price or refusal language. | Consider another term, not an imaginary price floor. |

### Unknown is a useful answer

A profile field is either **likely**, with a source, or **unknown**. Unknown values keep conservative opening terms. Repairing a buyer must-have or buying needed coverage is allowed separately; neither claims to know the supplier's motivation.

Code: arena_kit/docs.py and quote_extract.py read documents; negotiator/supplier_evidence.py and intelligence.py normalize them; supplier_profile.py applies explicit-statement rules; case_evidence.py scopes archived letters by author, supplier, part and date. Rules are mostly English. Broad OCR/locale robustness is not complete.

COMPONENTS 4-6

# A slightly higher price can be cheaper

A package is the price plus payment, quantity, duration, delivery and other contractual terms.

## The cost engine

The calculator adds purchase spend, uncovered need and fallback risk, financing, freight, late delivery, defects and chargebacks, tooling, index movement and index risk. It then checks every hard buyer requirement. For some strategic cases, Arena also adds a supplier-health penalty that needs private economics we do not have. Our estimate is therefore not universally exact.

| Observed P03 package | Quoted price | Actual / modeled whole cost |
|---|---|---|
| 24 months, 4,580 units/year, 30-day payment | EUR 430.15 | EUR 3,968,442.37 before the payment trade |
| Same coverage, 90-day payment | EUR 433.08 | EUR 3,942,826.22 accepted by Arena |
| Net effect | Price rose | Actual cost fell by EUR 25,616.15 |

## How candidate packages are created

The policy starts from the current quote, tests a few bounded opening prices, and keeps other terms unchanged unless a supported trade, buyer requirement or coverage benefit justifies a change. The generator evaluates allowed combinations. The safety selector sends only **one** complete, useful package and records the alternatives privately.

### Three gates before a proposal leaves

**1.** Every required field is present and allowed. **2.** Every buyer must-have holds. **3.** Modeled total cost stays below the reservation and available safe-offer ceiling. Previously sent exact packages are filtered. These gates do not guarantee the supplier will accept or preserve its previous offer.

## Two useful definitions

**Reservation** is our maximum acceptable whole cost, derived from the buyer's alternative or fallback. **Index clause** means the price follows a named market index. Its **deadband** is the movement ignored before an adjustment applies; a larger band can reduce the buyer's price exposure.

Code: arena_kit/cost.py is the published starter-kit arithmetic; negotiator/evaluator.py adds validation and explanations; package_policy.py chooses permitted changes; packages.py enumerates candidates; proposal_safety.py applies final limits. Unknown relationship cost remains a limitation even when every modeled gate passes.

COMPONENTS 7-10

# What happens after each reply?

The unified agent coordinates small modules; it does not ask a language model to calculate the deal.

| Situation | Action in plain language | Main component |
|---|---|---|
| Terms incomplete or contradictory | Ask a specific question. Do not guess missing values or silently waive a requirement. | dialogue_recovery / unified |
| A useful valid package exists | Send one explicit conditional proposal, priced by total cost. | package_policy / packages |
| Supplier changes the terms after our ask | Cost the entire change. A bounded premium can buy a useful term back. | term_repair |
| Price stalls or supplier invites a trade | Try one costed non-price exchange, retaining at least half the modeled benefit. | response_signal / response_trades |
| A cheaper current secondary sheet exists | Repeat its exact terms and request a primary standing offer; do not accept its secondary ID. | supplier_alternatives |
| An experiment made the offer worse | Request the earlier quoted package once. Ask for fresh standing confirmation. | offer_restoration (new) |
| Supplier requests support for our argument | Send eligible evidence or honestly say the ask is commercial. No invented benchmark. | buyer_arguments |
| Little benefit remains, final, or no moves | Accept the cheapest valid current offer under reservation, or walk away. | unified + runner |

The opening aspiration stays anchored to the original quote, not later concessions. Threshold-based stopping normally waits for two buyer messages. Final/deadline handling overrides exploration. These are deterministic heuristics, not proven optimal settings. Clarification is capped at two recovery/trade questions per thread.

COMPONENTS 11-13

# The plumbing matters as much as the strategy

An RFQ means several suppliers competing for one contract. Only one global acceptance closes it.

| Layer | Responsibility | Important boundary |
|---|---|---|
| CLI / agent factory | Load configuration and choose unified or a comparison agent. | supplier remains the default; unified is explicit. |
| HTTP client | Authenticate, pace calls, use 180-second HTTP timeout and retry the same move ID. | A network retry must not create a second move/session. |
| Runner | Download attachments, send supplier messages concurrently and close before the wave cap. | Parallel sends do not automatically keep idle rivals alive. |
| Validation harness | Freeze source, save selection/seed, run local or hosted cases, preserve decisions and results. | No replacement runs or score-based case filtering. |

### A real failure in the full benchmark

In P05 one supplier agreed early while another kept negotiating. The early supplier accumulated two timeouts; an uncompetitive third supplier accumulated three and withdrew. The winning thread had none. **Five timeouts is not an error-free run**, even though the final deal was valid.

## What the new closing check fixes

Immediately before closing, unified fetches current server state, including timeout-generated offer IDs, and re-evaluates the deal. It cannot simply attach an old price to a new ID. An unknown or over-budget updated offer is rejected. The original unsent closing decision is marked superseded.

## What it does not fix

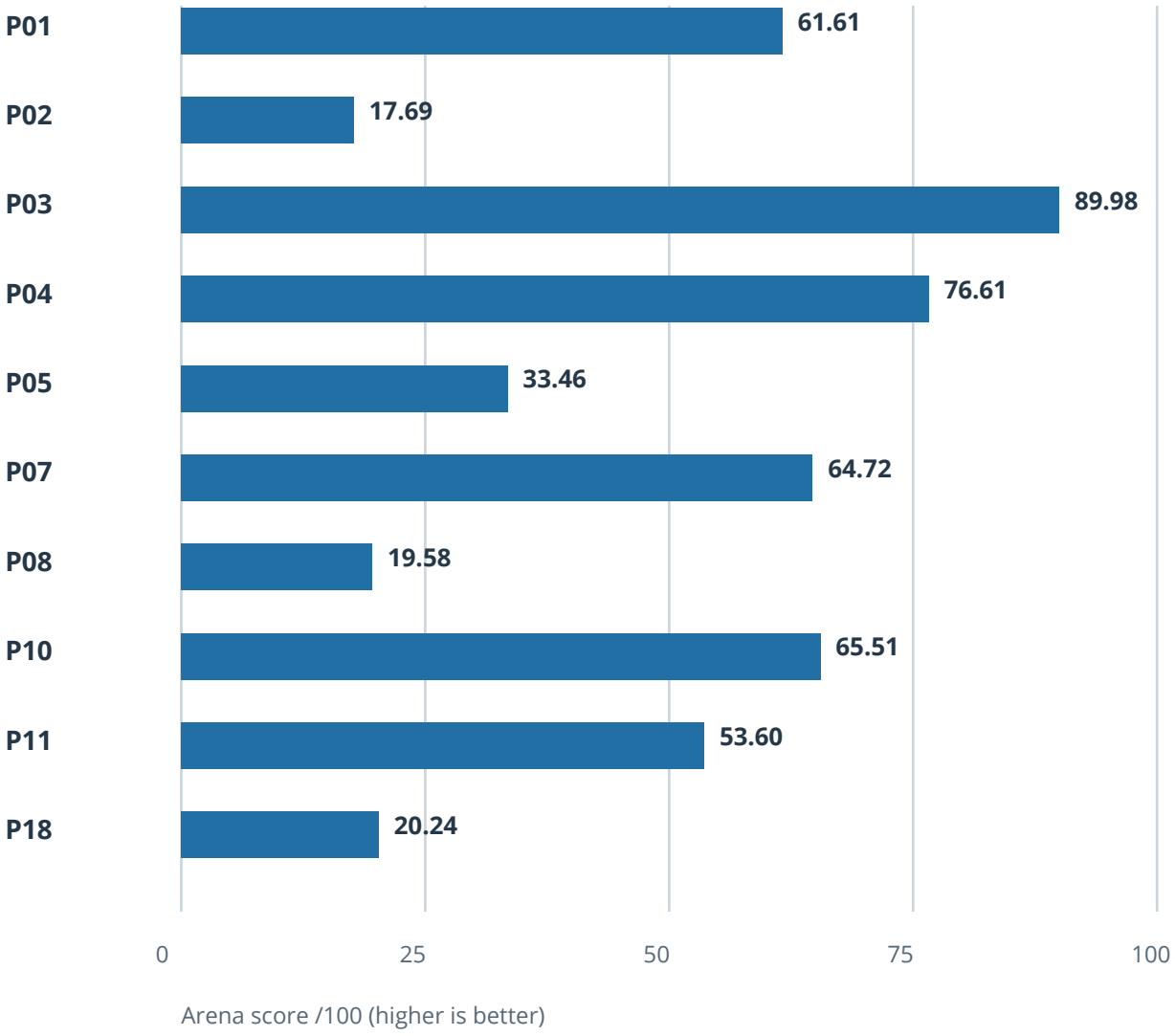
Reading state does not reset a supplier's 60-second clock. A full per-thread scheduler and an explicit policy for retiring uncompetitive suppliers are still needed. The API does not offer a documented 'park this supplier' action. Repeated keepalive emails would also consume rounds and affect the negotiation.

Code: arena_kit/cli.py, client.py and play.py; scripts/validate_mvp.py. The starter kit supplies much of the protocol and final-wave infrastructure; our contribution is its integration with the evidence-and-cost policy.

EXPERIMENT A

# All ten problems: 50.30 /100

One attempt per hosted case. Same frozen source. No replacement runs, no best-score selection.




## Coverage is stronger than the earlier random averages

This is a complete ten-case development benchmark. It includes the weaker cases that were absent from some earlier random batches. Do not compare 50.30 directly with a previous nine-run mean from a different mix and call the difference a regression or improvement.

10/10 deals met must-haves and actual reservation. Runner/contract-audit errors: 0. Recorded timeouts: 5, all in P05's inactive rivals. Modeled accepted cost matched Arena to the cent in 8/10 cases; the exceptions were P02 and P03's relationship component.

# Successes and weaknesses, side by side

These are observed mechanisms and limitations, not reconstructed supplier motives.

| Case / score | What happened and what it means |
|---|---|
| **P01** 61.61 | Sourced 14-day payment exchange led to 133.80. One purchase-order argument verified; acknowledgements no longer caused repeated evidence disclosure. |
| **P02** 17.69 | Clarification produced a complete copper-index alternative at 348.03. Confirmation worked, but unmodeled supplier-health cost added EUR 115,648.66. |
| **P03** 89.98 | A bounded higher-price / 90-day-payment exchange lowered actual cost. The oversized alternative was not chosen. Health-cost gap: EUR 259.63. |
| **P04** 76.61 | The 24-month package cut cost. Price history verified but did not move the supplier further. It made a final offer at 4.88; we accepted. |
| **P05** 33.46 | A 52-day quote failed the delivery limit; 42 days was repaired. Later payment raised price too far. Five idle-rival timeouts exposed stale-state risk. |
| **P07** 64.72 | 24-month coverage and a reciprocal counter reached 5.49. Supplier explicitly agreed; clean closing without further probing. |
| **P08** 19.58 | Supplier required indexation for 24 months. Its 5% deadband alternative at 5.72 was cheaper overall than the lower-price 0% deadband offer. |
| **P10** 65.51 | Opening volume was below the minimum. The sourced commitment package restored full coverage; final 31.80 / 56 days met the hard limits. |
| **P11** 53.60 | Quality improved from 6,400 to 1,500 ppm. Price ended at 46.00, worse than the earlier 45.50 outcome. More generous feasibility credit needs study. |
| **P18** 20.24 | The supplier insisted on indexation. It rejected a higher-price / better-deadband exchange, then made a final offer. We did not force the clause. |

All returned buyer and supplier messages, structured offers, statuses and argument results were reviewed, including all three P05 suppliers and their separate timeout records. A verified citation is useful evidence hygiene, not an automatic reward.

EXPERIMENT B

# Reflect, change, test, then run three

The three cases were selected uniformly without replacement after the second local gate.

| Case | Before /100 | After /100 | Change |
|---|---|---|---|
| P01 | 61.61 | 61.61 | +0.00 |
| P05 | 33.46 | 35.21 | +1.76 |
| P18 | 20.24 | 20.24 | +0.00 |
| Mean | 38.44 | 39.02 | +0.59 |

## Changes before these three runs

**Recovery:** one request to renew a better prior primary quote after an unhelpful experiment; no stale-ID acceptance. **Language:** only first-hand prospective trade invitations; quoted or negated requests cannot trigger document disclosure. **Responsiveness:** answer genuine evidence requests even when our move is a question. **Closing:** fetch current remote state, then recheck safety.

## P01: 61.61 /100

Unchanged at 133.80 / 14-day payment. The source-backed cash trade and one verified history argument still worked; no repeated history disclosure. The new pre-close state refresh completed normally.

## P05: 35.21 /100

The new restoration request was exercised: the supplier renewed 32.50 / 45 days / 42-day delivery, saving EUR 1,442 versus the benchmark deal. Score rose by 1.76 points. No timeouts this time, but replies were far faster; this does not prove scheduling is fixed.

## P18: 20.24 /100

Unchanged: the supplier again rejected the better index band and made a final offer. The agent accepted the valid current package. The changes did not unlock additional commercial value here.

Follow-up runner/contract-audit errors: 0; recorded timeouts: 0. The mixed 13-run mean is 47.70, but it combines versions and repeats three cases, so it is not the headline benchmark. Practice cases are not an unseen holdout; three observations cannot establish a general causal improvement.

NEXT DECISIONS

# Protect what works. Test what is uncertain.

Do not confuse architectural completion, commercial strength and final-event readiness.

| Priority | Next action | Why / proof required |
|---|---|---|
| 1 - Reliability | Add independent per-thread deadline handling and deliberate RFQ retirement/closing decisions. | A ten-case run exposed five idle-rival timeouts. Stress-test slow and uneven replies; fresh closing alone is not enough. |
| 2 - Strategy | Run a predeclared matched comparison of concession / feasibility-credit settings. | P11 regressed. Freeze one change per arm, keep every outcome, use varied synthetic inputs, never fit hidden best prices. |
| 3 - Model limits | Expose uncertain relationship cost in decision diagnostics; use genuine disclosed economics only. | P02's missing component is material. Do not reverse-engineer private costs from scores or claim a guaranteed true-cost ceiling. |
| 4 - Evidence | Harden mixed/scanned PDFs, locale ambiguity and conflict handling in an isolated module. | Current parsing and mostly English rules are narrow. Add fixtures and failure diagnostics before broadening inference. |
| 5 - Deployment | Rehearse wave deadlines and resume behavior on the final machine; prepare release data and submission. | A good practice score does not prove eight-minute wave readiness or satisfy the repo/hash/slide submission requirements. |

### Keep these protections

Unknown means unknown. No case-ID tactics or learned supplier floors. One clear proposal, complete terms, hard buyer limits, costed reciprocal trades and exact standing-offer acceptance. Keep supplier, packages and advanced runnable; do not make unified the default from these thirteen observations.

The optional language-model (LLM) agent and evidence-priority adapter remain separate. The evaluated unified path is deterministic: it uses no LLM calls. Final-event operations described in the bundled submission guide were not executed by this task.

AUDIT TRAIL

# Thirteen sessions, two frozen versions

Replay links open the actual hosted conversations. Local files retain the complete decisions and cost audits.

| Run | Score /100 | Replay ID (clickable) |
|---|---|---|
| 10-case / P01 | 61.61 | 1e995020ca5f4f6f |
| 10-case / P02 | 17.69 | da4c937cb970d8fb |
| 10-case / P03 | 89.98 | 278b82835e069cdc |
| 10-case / P04 | 76.61 | 1fade1b5c9678110 |
| 10-case / P05 | 33.46 | ad91c4cbafd812d1 |
| 10-case / P07 | 64.72 | 229583c08bb6c760 |
| 10-case / P08 | 19.58 | a91a2355894277f1 |
| 10-case / P10 | 65.51 | a31120c6c6816049 |
| 10-case / P11 | 53.60 | 413fd2afb055a2f2 |
| 10-case / P18 | 20.24 | cd6b0e6cddb93461 |
| Follow-up / P01 | 61.61 | 191a2683bd93fb32 |
| Follow-up / P05 | 35.21 | 3a8f872fd7caf8b0 |
| Follow-up / P18 | 20.24 | 985c50501cf01ff5 |

**Source fingerprints** (working-tree versions, not committed releases)

Ten-case: 7e618c0819fff26779815ce74e75759cd3034208cf6a89b4c6be5bb92de44400

Follow-up: c41b464d4c12b9aa4a77b391265ddb6ff5a1b8003df779b85bdc4a4183559511

Git base: cb01512947a07d98ae020af3014d968d21ef06cd
Random follow-up seed: 2742409430. Selected before opening sessions; no replacement runs.
Local tests: 288 before benchmark; 305 before follow-ups; each with 1 skipped and 15 subtests passed. All four agents' S01-S03 scores stayed unchanged. Advanced remains stronger on those scripted cases; they do not model real supplier behavior.

**Sources and files:** bundled brief/brief.md, brief/api.md, brief/submission.md; README.md, MVP_INTELLIGENCE.md, MVP_EVALUATOR.md, MVP_ROUND3_REPORT.md, MVP_CORE_CHARACTERISTICS.md; inspected implementation and tests. Run evidence: output/validation/full-suite-10-hosted/ and full-suite-followup-3-hosted/. Plan: MVP_FULL_SUITE_PLAN.md. A source hash is not a dependency or server-behavior lock. Secrets are not included in this report.